Data Analytics (DA)

Module End Assignment:

Data Cleaning, Analysis & Visualization with Excel"Tasks

1. **<u>Data entry, import, and basic organization using tables:-</u>**

Creating a Table Name as **Product_DB, Customer_DB, Store_DB and Sales_DB**

A	B	C	D	E
Product ID	Product Nam	Product Name_Standardization	Brand Nam	Price ($)
28-JAN-US	laptop	Laptop	Dell	1000
15-FEB-US	Sneakers	Sneakers	Nike	80
03-MAR-US	Coffee Maker	Coffee maker	Keurig	130
11-APR-US	smartphone	Smartphone	Samsung	900
22-MAY-US	Backpack	Backpack	North Face	70
07-JUN-UK	Headphones	Headphones	Sony	
19-JUL-UK	T-shirt	T-shirt	Adidas	30
23-AUG-UK	Blender	Blender	Ninja	90
05-SEP-UK	Tablet	Tablet	Apple	500
14-OCT-UK	Hiking Boots	Hiking boots	Timberland	130
17-JUN-IN	Laptop	Laptop	HP	950
25-NOV-AU	Sneakers	Sneakers	Adidas	90
08-DEC-DE	Coffee Maker	Coffee maker	Nespresso	120
18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150
16-APR-ES	headphones	Headphones	Bose	250
21-AUG-CA	Laptop Bag	Laptop bag	Samsonite	50
20-AUG-CN	Smartwatch	Smartwatch	Huawei	160
27-JAN-IT	laptop	Laptop	Asus	980
01-MAR-UK	Sunglasses	Sunglasses	Oakley	150

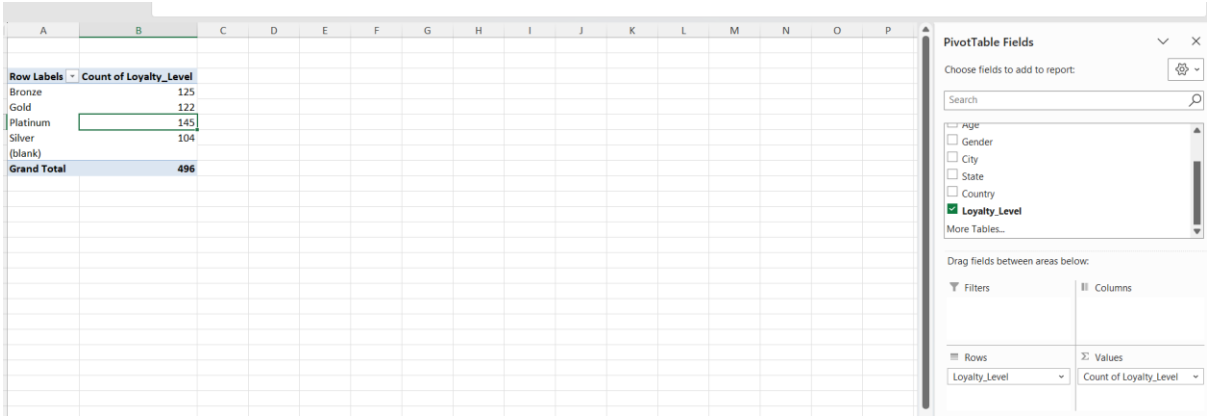
2. **Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data:-**

Data Cleaning:-

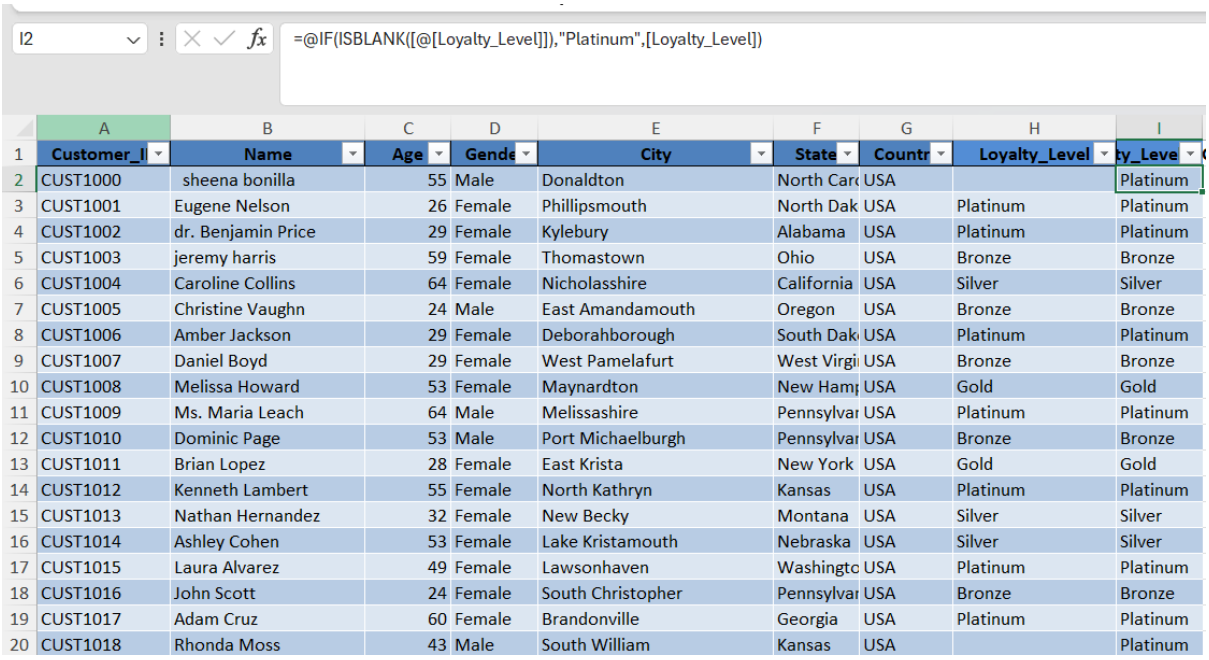
1. **Customer Table:-**

- Handling Missing Values in the the 'Loyalty level' column in Table Customer_Dim Using Pivot table and find the Top one and using Isblank function to fill blank

column.



Row Labels	Count of Loyalty_Level
Bronze	125
Gold	122
Platinum	145
Silver	104
(blank)	
Grand Total	496



I2 =@IF(ISBLANK([@[Loyalty_Level]]),"Platinum",[Loyalty_Level])

	A	B	C	D	E	F	G	H	I
1	Customer_I	Name	Age	Gende	City	State	Countr	Loyalty_Level	ty_Leve
2	CUST1000	sheena bonilla	55	Male	Donaldton	North Car	USA		Platinum
3	CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dak	USA	Platinum	Platinum
4	CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum	Platinum
5	CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze	Bronze
6	CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver	Silver
7	CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze	Bronze
8	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dak	USA	Platinum	Platinum
9	CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virgi	USA	Bronze	Bronze
10	CUST1008	Melissa Howard	53	Female	Maynardton	New Hamp	USA	Gold	Gold
11	CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvar	USA	Platinum	Platinum
12	CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvar	USA	Bronze	Bronze
13	CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold	Gold
14	CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum	Platinum
15	CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver	Silver
16	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver	Silver
17	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washingto	USA	Platinum	Platinum
18	CUST1016	John Scott	24	Female	South Christopher	Pennsylvar	USA	Bronze	Bronze
19	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum	Platinum
20	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA		Platinum

- Correcting Inconsistent Data present in the " Name" column in Customer Table. Using function Proper and Trim to change first letter in the product name in upper case and removes unwanted space.

C2 =PROPER(TRIM(B2))

	A	B	C	D	E	F
1	Customer_Id	Name	Name_Clean	Age	Gende	City
2	CUST1000	sheena bonilla	Sheena Bonilla	55	Male	Donaldton
3	CUST1001	Eugene Nelson	Eugene Nelson	26	Female	Phillipsmouth
4	CUST1002	dr. Benjamin Price	Dr. Benjamin Price	29	Female	Kylebury
5	CUST1003	jeremy harris	Jeremy Harris	59	Female	Thomastown
6	CUST1004	Caroline Collins	Caroline Collins	64	Female	Nicholasshire
7	CUST1005	Christine Vaughn	Christine Vaughn	24	Male	East Amandamouth
8	CUST1006	Amber Jackson	Amber Jackson	29	Female	Deborahborough
9	CUST1007	Daniel Boyd	Daniel Boyd	29	Female	West Pamelafurt
10	CUST1008	Melissa Howard	Melissa Howard	53	Female	Maynardton
11	CUST1009	Ms. Maria Leach	Ms. Maria Leach	64	Male	Melissashire
12	CUST1010	Dominic Page	Dominic Page	53	Male	Port Michaelburgh
13	CUST1011	Brian Lopez	Brian Lopez	28	Female	East Krista

2. Product Table:-

- ● Identified the missing values in the Cost and Stock column used Isblank, Average and Round functions to fill the blanks cells.

G2 =ROUND(IF(ISBLANK([@Cost]),AVERAGE([Cost]),[@Cost]),0)

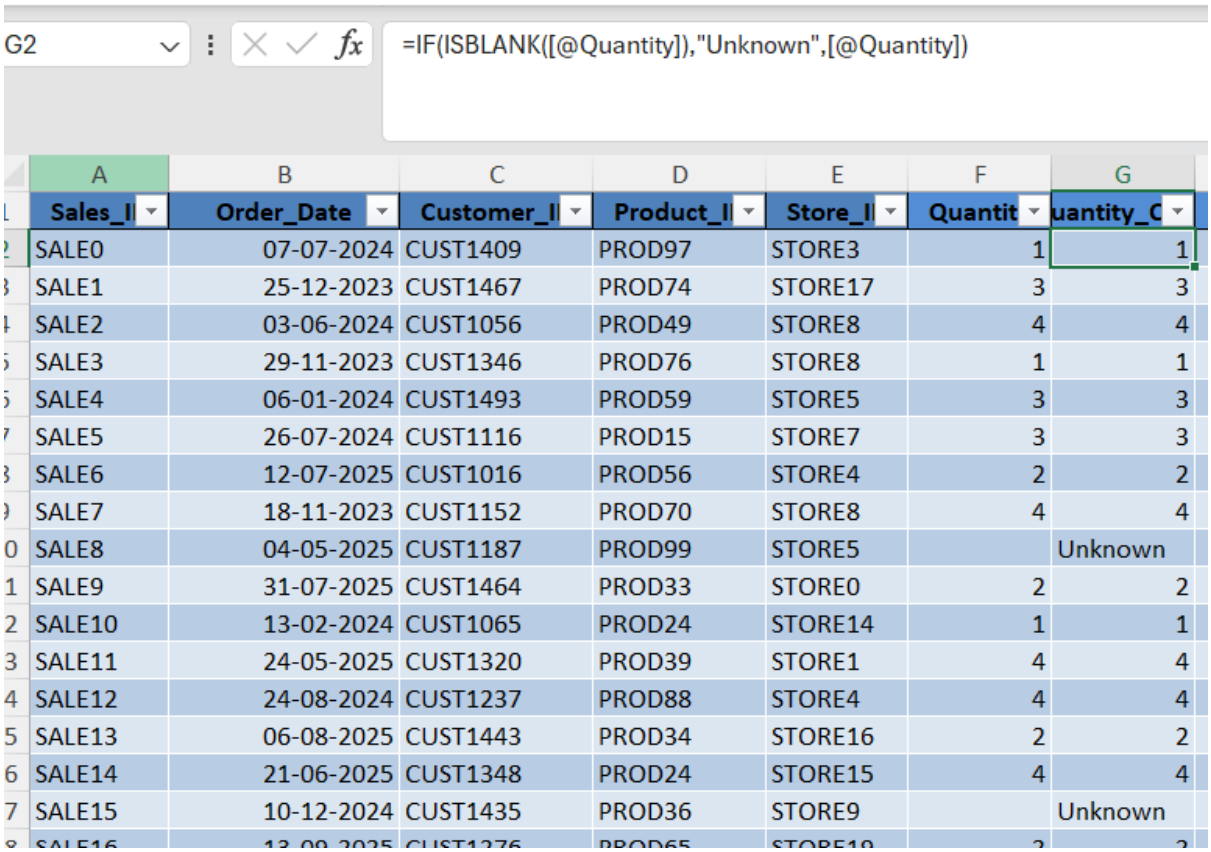
	A	B	C	D	E	F	G
1	Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Cost_Clea
2	PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	112
3	PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	343
4	PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		254
5	P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	365
6	PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	306
7	PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	261
8	PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	313
9	PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	219
10	PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	439
11	PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	496
12	PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	299
13	PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	449
14	PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	116

I2 =IF(ISBLANK([@Stock]),"0",[@Stock])

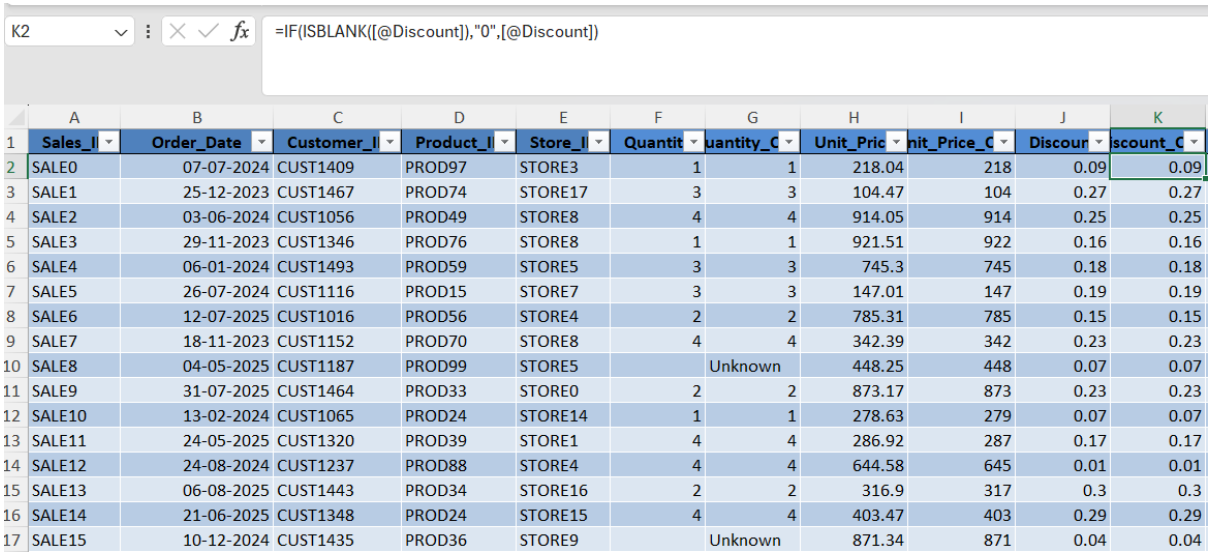
	A	B	C	D	E	F	G	H	I
1	Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Cost_Clea	Stock	Stock_Clean
2	PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	112	416	416
3	PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	343	181	181
4	PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		254	352	352
5	P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	365	452	452
6	PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	306	487	487
7	PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	261	59	59
8	PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	313	350	350
9	PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	219	170	170
10	PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	439	142	142
11	PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	496	323	323
12	PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	299	308	308
13	PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	449		0
14	PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	116	460	460
15	PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	242	453	453

3. Sales Table:-

- Using isblank function to fill the quantity and discount colums.



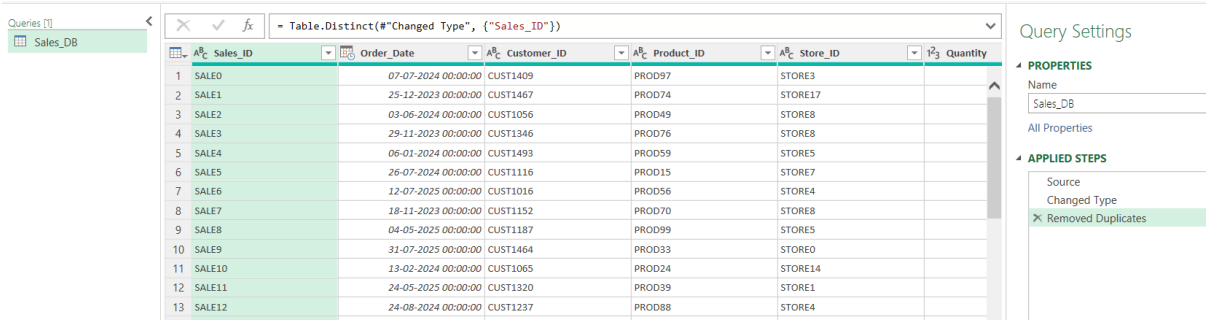
	A	B	C	D	E	F	G
1	Sales_I	Order_Date	Customer_I	Product_I	Store_I	Quantit	uantity_C
2	SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	1
3	SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	3
4	SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	4
5	SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	1
6	SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	3
7	SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	3
8	SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	2
9	SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	4
10	SALE8	04-05-2025	CUST1187	PROD99	STORE5		Unknown
11	SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	2
12	SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	1
13	SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	4
14	SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	4
15	SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	2
16	SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	4
17	SALE15	10-12-2024	CUST1435	PROD36	STORE9		Unknown



	A	B	C	D	E	F	G	H	I	J	K
1	Sales_I	Order_Date	Customer_I	Product_I	Store_I	Quantit	uantity_C	Unit_Pric	nit_Price_C	Discoun	scount_C
2	SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	1	218.04	218	0.09	0.09
3	SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	3	104.47	104	0.27	0.27
4	SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	4	914.05	914	0.25	0.25
5	SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	1	921.51	922	0.16	0.16
6	SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	3	745.3	745	0.18	0.18
7	SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	3	147.01	147	0.19	0.19
8	SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	2	785.31	785	0.15	0.15
9	SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	4	342.39	342	0.23	0.23
10	SALE8	04-05-2025	CUST1187	PROD99	STORE5		Unknown	448.25	448	0.07	0.07
11	SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	2	873.17	873	0.23	0.23
12	SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	1	278.63	279	0.07	0.07
13	SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	4	286.92	287	0.17	0.17
14	SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	4	644.58	645	0.01	0.01
15	SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	2	316.9	317	0.3	0.3
16	SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	4	403.47	403	0.29	0.29
17	SALE15	10-12-2024	CUST1435	PROD36	STORE9		Unknown	871.34	871	0.04	0.04

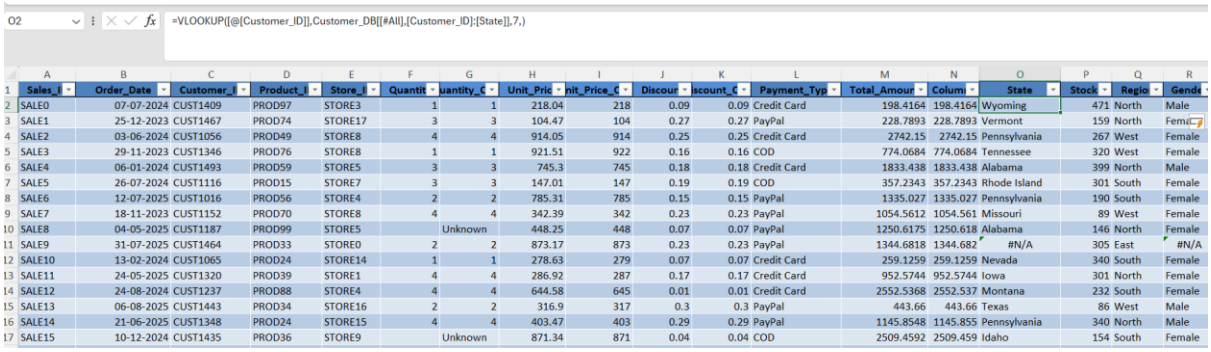
3. Remove the column that contains irrelevant data if it exists:-

- Removing the duplicates in the sales table using power query.



4. **Application of Excel formulas for calculations, summaries, and logical/text operations:-**
   - Customer Table – Used logical function if
   - Product Table – Used logical function if and mathematical function round
   - Sales Table – Used logical function if and mathematical function round and average.

5. **Use the XLOOKUP/VLOOKUP functions to combine the tables:-**
   - Using Vlookup to combine all the tables.



6. **Use of Pivot Tables for advanced summarization:-**

   - Creating Pivot Tables for various analysis

Row Labels	Sum of Unit_Price_Clean	Count of Sales_ID
East	6760	11
North	9565	18
South	6640	13
West	2411	4
Grand Total	**25376**	**46**

Row Labels	Count of Total_Amount	Sum of Total_Amount
Female	24	25817.876
Male	22	23559.8084
Grand Total	**46**	**49377.6844**

Row Labels	Sum of Total_Amount
Oklahoma	59386.5411
New Jersey	65274.9563
New Hampshire	69579.9252
Kansas	58052.2846
Florida	60406.6758
Grand Total	**312700.383**

7. **Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak :-**

Descriptive statistics	Values	Functions Used
Total Sales	2170907	SUM
Min Sales	18.459	MIN
Max Sales	3908.243	MAX
Mean	1087.629	AVERAGE
Median	868.7182	MEDIAN
Total Sales more than 500 amount	1419	COUNTIF
Standard Deviation	820.1258	STDEV.S

8. **Data visualization: create at least three chart types, build an interactive dashboard, and use basic PivotCharts:-**

